

Pham Quang Huy

Software Engineer (AI Systems & Web Platforms)

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SUMMARY

Software Engineer with nearly 3 years of experience building real-time AI camera systems, backend services, and enterprise web platforms. Experienced with C++17, NVIDIA DeepStream, Triton, GStreamer, Kafka, Redis, Docker, Linux, Java backend, and React/Next.js. Interested in Computer Vision, face recognition, video analytics, and high-performance streaming systems.

EDUCATION

FPT University

Software Engineer | 09/2019 – 12/2023

GPA: 3.5/4.0 – Very Good

FPT Software Academy

Korean Bridge Software Engineer | 09/2022 – 12/2023

GPA: 9.0/10

CERTIFICATIONS & SCHOLARSHIPS

- OCP Java SE 17 Developer (1Z0-829) – Oracle
- TOPIK 4 - National Institute for International Education
- 100% Scholarship - Korean Bridge SE program, FPT Academy

SKILLS

- **Programming Languages:** Java, C++17, TypeScript/JavaScript, Erlang, Basic Python
- **Computer Vision / Streaming:** NVIDIA DeepStream, Triton Inference Server, GStreamer, RTSP
- **Backend & Middleware:** Spring Boot, NestJS, Quarkus, OSGi, Kafka, ActiveMQ
- **Databases & Vector Search:** PostgreSQL, MySQL, MongoDB, Qdrant, Redis, Mnesia
- **Frontend Frameworks & UI:** ReactJS, Next.js, AngularJS, Tailwind CSS, Bootstrap, MUI
- **DevOps & Systems:** Linux, Docker, Jenkins, GitHub Actions, Basic AWS
- **Tools:** Git, SVN, SonarLint, Postman, DBeaver, Power BI, SoapUI
- **Languages:** English (Intermediate), Korean (TOPIK 4)

PROJECTS

Camera AI Surveillance System | C++/AI Backend Engineer

FPT Software | 06/2025 – 01/2026

Technologies: C++17, NVIDIA DeepStream, Triton, GStreamer, Kafka, Redis, Qdrant, Docker, Golang, Linux
Developed a real-time AI camera system for face recognition, face embedding search, and metadata streaming on **NVIDIA** GPU-based **Linux** servers.

- Built **C++17** video processing modules using **NVIDIA DeepStream** and **GStreamer** to process continuous **RTSP** camera streams in real time.
- Integrated **Triton Inference Server** and local fallback recognition paths for GPU-accelerated face recognition workflows.
- Built event and metadata streaming pipelines using **Kafka**, **Redis**, and **ZMQ** for recognition results and runtime status updates.

- Implemented lightweight video analytics logic based on detection/tracking inputs for server-side and edge-oriented deployment paths.
- Used **Qdrant** to store face embeddings, perform similarity search, and support identity matching.
- Debugged and deployed services in **Docker**-based **Linux** environments on **NVIDIA GPU** servers.

Smart Connect | Backend/Middleware Developer

FPT Software – LG CNS Client Project | 05/2024 – 01/2025

Technologies: Java, OSGi, ActiveMQ, C++ Native Library Integration, Docker, Linux

Built **Java**-based middleware services for integrating biometric authentication devices, including facial recognition and fingerprint recognition, with enterprise platforms through real-time messaging.

- Designed middleware modules using **Java OSGi** to integrate biometric devices with enterprise systems.
- Implemented **ActiveMQ**-based asynchronous messaging for device events, authentication results, and status updates.
- Integrated native **C++** libraries with **Java** services for biometric processing workflows.
- Debugged message routing, device synchronization, and deployment in **Docker/Linux** environments.

Telecommunications Core Routing System | Backend Developer

FPT Software – Viettel High Tech (Onsite) | 03/2026 – 05/2026

Technologies: Erlang, Java, Quarkus, Mnesia, MySQL, Linux

Worked on a high-concurrency telecom core routing system for real-time communication flows and routing business logic.

- Implemented and optimized routing business logic using **Erlang** and **Java Quarkus**.
- Managed distributed runtime data with **Mnesia** and persistent business data with **MySQL**.
- Simulated **Linux**-based testing environments to validate routing scenarios and fault-tolerance behavior.

SecuxperDP | Fullstack Developer

FPT Software – LG CNS Client Project | 01/2025 – 06/2025

Technologies: Java Spring Boot, Next.js, PostgreSQL, MongoDB, Jenkins, SonarLint, Docker, JWT

Contributed to a full-stack enterprise platform, focusing on backend services, frontend interfaces, and authentication.

- Developed **Java Spring Boot** backend services, including user management and **JWT** authentication.
- Built **Next.js** frontend features and integrated secured RESTful APIs with role-based access control.
- Worked with **PostgreSQL** and **MongoDB** for relational business data and configuration storage.
- Supported CI/CD workflows with **Jenkins**, **Docker**, and **SonarLint**.

Honda Digital User Manual | Frontend Developer

FPT Software – Honda Client Project | 02/2026 – 03/2026

Technologies: ReactJS, NestJS, Tailwind CSS

- Rebuilt legacy digital manual pages into a responsive **React**-based interface.
- Developed **NestJS** APIs to serve structured documentation content and support frontend integration.

IDP Plus | Fullstack Developer

FPT Software – LG CNS Client Project | 12/2023 – 05/2024

Technologies: Java Spring Boot, AngularJS, PostgreSQL, MongoDB, Redis, Jenkins, SonarLint

- Built internal enterprise features for employee and department data management.
- Developed **Java Spring Boot** APIs and **AngularJS** admin interfaces.

Project KPI Analytics Platform | Data Analyst / BI Developer

FPT Software | 01/2026 – 02/2026

Technologies: Power BI, DAX, Power Query, Data Lake Integration

- Built **Power BI** dashboards, semantic models, and **DAX** measures to monitor departmental project KPIs.
- Integrated and transformed Data Lake sources with **Power Query** to support real-time reporting and performance analysis.